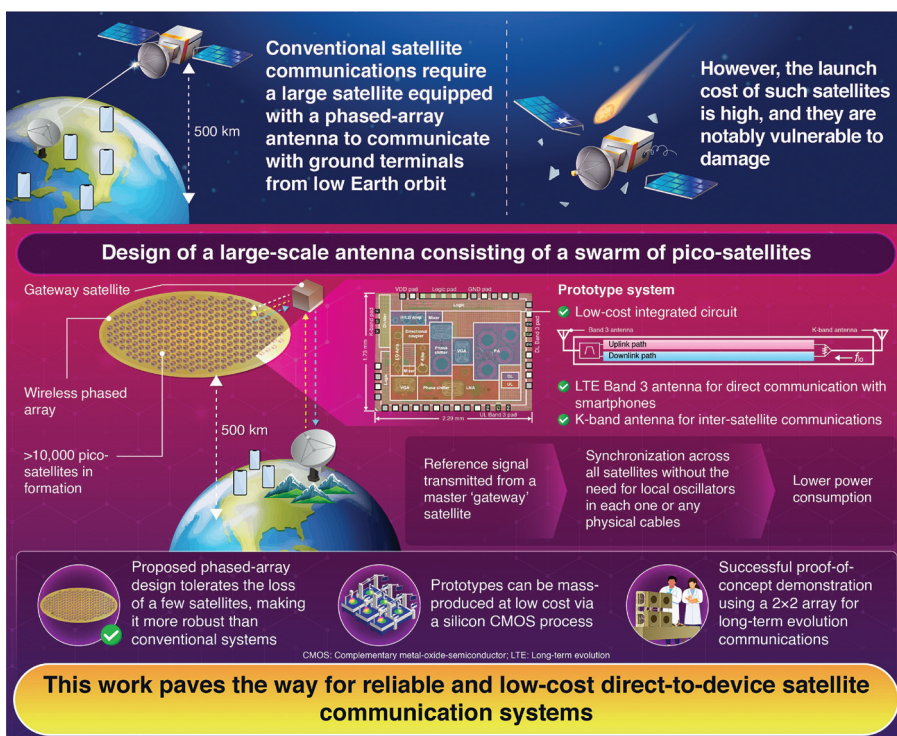


Pico-satellites to connect remote areas

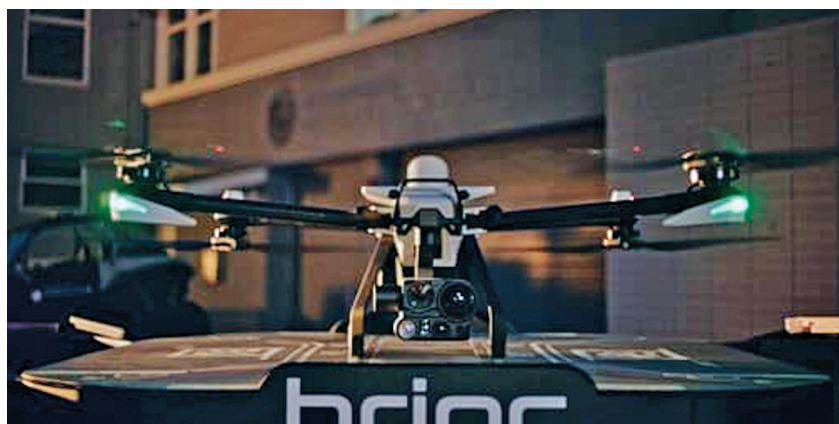


This work proposes an innovative system in which the elements of a phased-array antenna are distributed among thousands of extremely small satellites, avoiding the costs and risks associated with relying on a single large satellite (Credit: Institute of Science Tokyo)

Swarms of pico-satellites could soon transform mobile connectivity by enabling direct communication between smartphones and space-based networks. The approach uses thousands of tiny satellites working collectively as a large antenna, improving signal reach and reducing reliance on terrestrial infrastructure. Developed through research conducted at the Institute of Science, Tokyo, Japan, the system leverages coordinated satellite arrays to boost coverage and resilience. The innovation could expand connectivity to remote regions and enhance emergency communications. It builds on advances in low-cost, miniaturised satellite technologies, driving next-generation global networks.

Drone promises autonomous emergency response

A new 'Guardian' drone system promises continuous, always-on emergency response by eliminating downtime between missions. The platform enables 24/7 deployment through an automated station that swaps batteries and payloads without human intervention, ensuring rapid redeployment. Developed by a team in Seattle, the drone integrates satellite connectivity, long-range operation, and advanced imaging, including thermal cameras and high-zoom video for real-time situational awareness. The system could transform public safety by delivering medical supplies and conducting surveillance before first responders arrive, marking a shift toward autonomous, drone-led emergency-response infrastructure.



BRINC Guardian lands in Guardian Station after a successful mission (Credit: <https://interestingengineering.com>)